Data Warehouses and Data Mining Systems

Laboratory 4: Report content

1 Introduction

The data warehouse physical model is in fact a relational model associated with an actual database system. It consists of relations (i.e. tables) and some additional information, such as data types and constraints for example.

2 Database management system

The first step in building a data warehouse system is choosing the right database management system and other components (ETL, cubes etc.). Some vendors provide a full package of all required elements (e.g. Microsoft SQL Server + SSIS + SSAS + SSRS), while some cover only the necessary minimum (e.g. MySQL database). In the latter case one should look for additional tools (frameworks, programs etc.) in order to implement the whole data warehouse.

3 Physical model

The physical model of the data warehouse should be developed basing on the logical model defined previously. It should provide the following information:

1. Data types for each attribute;
2. Primary and foreign keys;
3. Any further technical details (data size for varchar, additional constraints etc.)

4 DDL SQL statements

Given a physical model, one should be able to compose DDL SQL statements that could be used to create real tables in the selected database management system. The DDL statements should be contained in the report too.